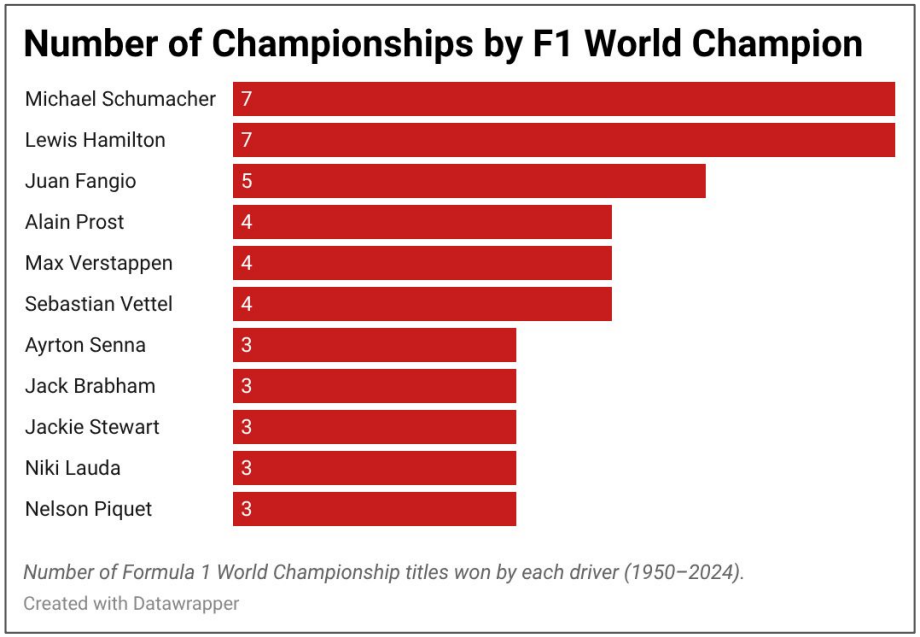
Data Visualization Pitch

Student Individual Assignment



What makes an F1 driver a World Champion?

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Research questions

RQ: What makes an F1 driver a World Champion?

- SQ1: Where do F1 World Champions come from?
- SQ2: Do F1 world champions win more races?
- SQ3: Does driving for the best team guarantee a world championship?

About Data

The dataset **Formula 1 World Championship (1950 - 2024)** (available here) provides exhaustive information about F1 races, drivers, constructors, qualifying, circuits, lap times, pit stops, and championships from the 1950 season through the 2024 season.
It is released under the license "CC0: Public Domain", allowing unrestricted use, making it fully suitable for academic purposes.

Despite its richness, the dataset presents some challenges regarding interpretability and comparability. The main one is that the F1 scoring system changed several times throughout history. The most significant revision occurred in 2010, making direct comparison between different eras potentially problematic. Additionally, sprint races were introduced in 2021, creating new opportunities for drivers to score points and slightly altering championship dynamics.

In order to address this issues, data cleaning and preprocessing steps were applied, including table merging, variable standardization, and temporal filtering.

A check for missing values revealed that all datasets used in the analysis are complete. The only null values appear in the *qualifying* table, specifically in the Q2 and Q3 variables, which is expected since not all drivers advance to later qualifying sessions. As these variables were not used in the analysis, no additional data imputation was required.

Methodology

Data Collection: All data was downloaded manually from the **Formula 1 World Championship (1950 - 2024)** dataset available on Kaggle.

Data Processing & Cleaning: Data processing and cleaning were performed using Python, with the pandas library. To ensure comparability, data was filtered to include only seasons from 2010 onward to ensure that all comparisons involving points and win rates (SQ2 and SQ3) were performed under a consistent scoring system. This filtering directly addressed the main comparability challenge identified in the dataset.

Data Transformation: Tables were merged and aggregated according to each research question:

- SQ1: summarizing champions by nationality
- SQ2: normalizing wins and points relative to season totals
- SQ3: aggregating drivers and constructor final standings

Use of AI tools: AI tools (ChatGPT) were used to support code structuring.

Data Visualization: All charts and maps were created using Datawrapper and Flourish. Visualization choices were guided by clarity, readability and alignment with the specific question.

Analytical Techniques: Analytical techniques included filtering, aggregation, normalization and ranking in order to extract meaningful insights.

Reproducibility: The entire analysis is fully reproducible. All scripts were written in python and rely on public datasets. Code repository: here.

Insights from the Data

The analysis reveals a strong concentration of success among a very small group of elite drivers: while most drivers have few or no victories, only a handful consistently achieve poles, wins, and podiums.

- Lewis Hamilton dominates in absolute terms, holding records for pole positions (107), wins (105), and podiums (202), illustrating how long-term competitiveness amplifies career totals.
- When normalized by participation, Juan Fangio emerges as the most efficient driver historically, with the highest win rate (41.38%) and podium rate (60.34%), showing that dominance can also be measured through relative performance rather than raw numbers.
- While high DNF rates can limit success (like Andrea de Cesaris' 205 DNFs or Yannick Dalmas' 100% DNF rate), low DNF rates support championship consistency, as exemplified by Lewis Hamilton's 12.36%, highlighting the crucial role of reliability in shaping careers."
- At the team level, Ferrari's 16 world titles confirm that sustained constructor performance is a key factor of driver success, reinforcing the importance of the driver-car combination, followed by McLaren and Williams with 9 titles each, Mercedes with 8, and Red Bull with 6.

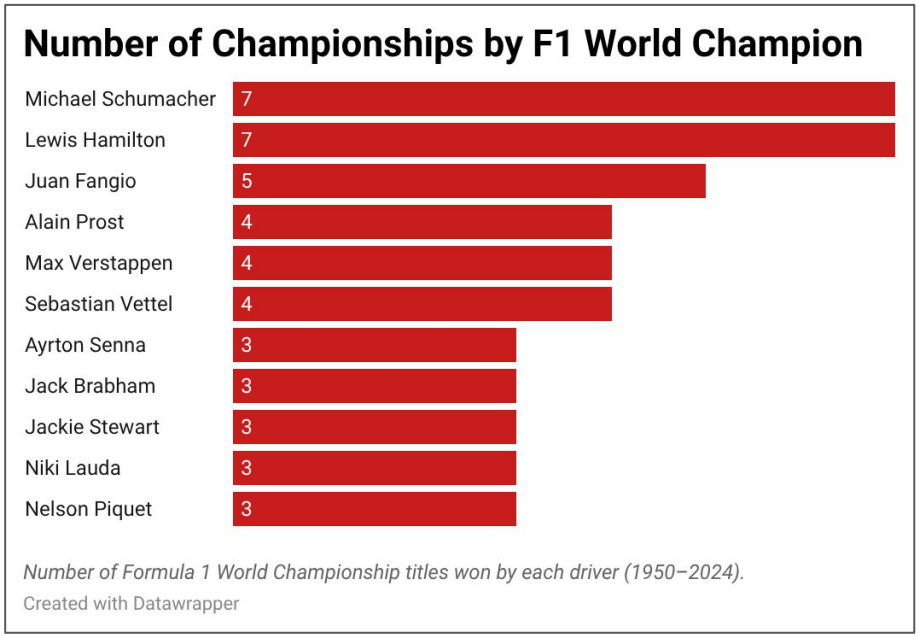
Key takeaway: Becoming a World Champion is not determined by a single factor, but by a combination of individual talent, consistency, reliability, and team strength.

Notes: Rates and comparisons consider only drivers with at least 10 race starts to avoid distortions caused by very short careers.

Only the Elite Compete

Historically, most champions have won multiple titles, demonstrating that once a driver reaches the top, they often remain highly competitive for several seasons. Michael Schumacher and Lewis Hamilton lead with 7 titles each, followed by Juan Fangio with 5 titles. A bar chart showing the number of championships per driver clearly illustrates this dominance.

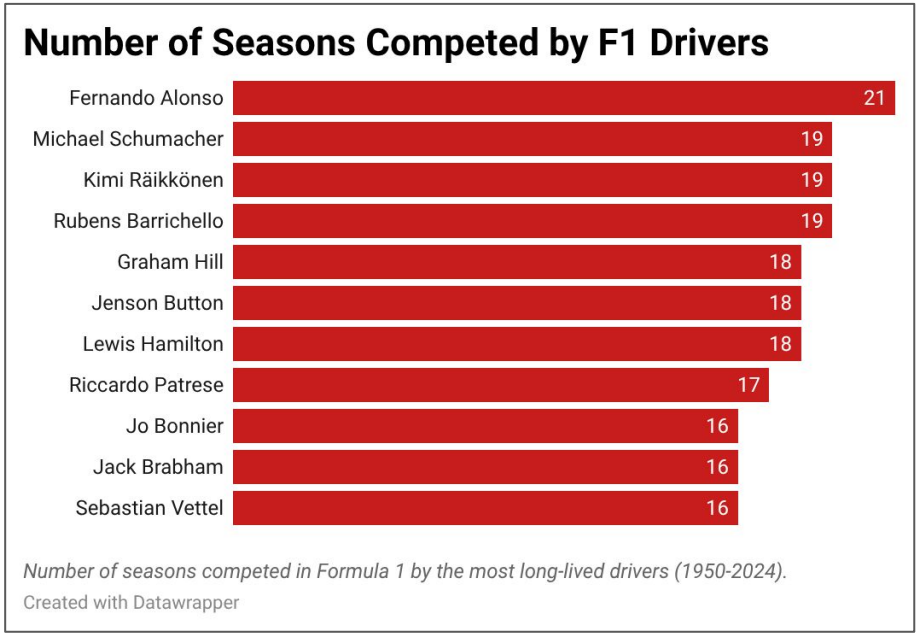
Focusing on the period from 2010 onward, only four drivers (Lewis Hamilton, Sebastian Vettel, Max Verstappen and Nico Rosberg) have claimed championships, showing that titles have been concentrated among a small group of elite drivers. The fact that so few drivers win multiple titles underscores the elite performance needed to become a World Champion.



Longevity Turns Drivers Into Legends

Another interesting fact is the drivers' career longevity. Some drivers have competed for remarkably long periods. Fernando Alonso leads with 21 season, followed by Michael Schumacher, Kimi Räikkönen and Rubens Barrichello, each with 19 seasons. A bar chart showing the number of seasons per driver highlights the most experienced competitors, providing a different perspective on success beyond championship titles.

While longevity alone doesn't guarantee championships, it underscores the value of consistency in achieving success.



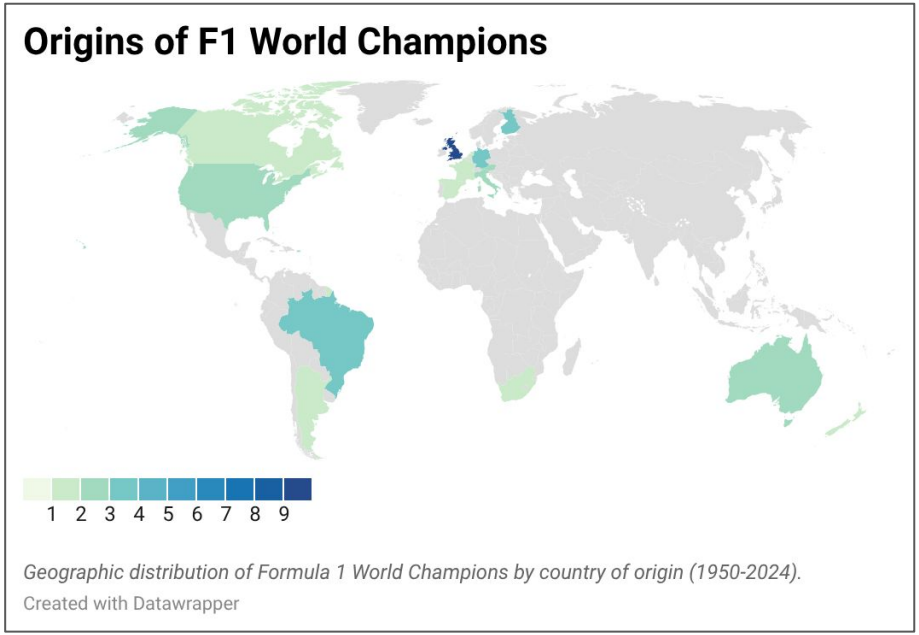
Europe is Home of Champions

SQ1: Where do F1 world champions come from?

The map shows the number of F1 world champions per country. It is important to notice that drivers who won multiple titles are counted once, in order to focus on the countries rather than the total number of championships.

The Great Britain clearly dominates, having produced more world champions than every other country, with 10 different winners. Germany, Brazil and Finland follow with 3 champions each, while Italy, Austria and the United States have produced 2 champions apiece. Other countries - including Argentina, Canada, France, the Netherlands, South Africa, Spain, New Zealand - have each celebrated a single world champion.

Overall, the map highlights Europe as the historical and technical heart of motorsport, showing that most success in the sport has been concentrated there, with only a few notable exceptions in other continents.

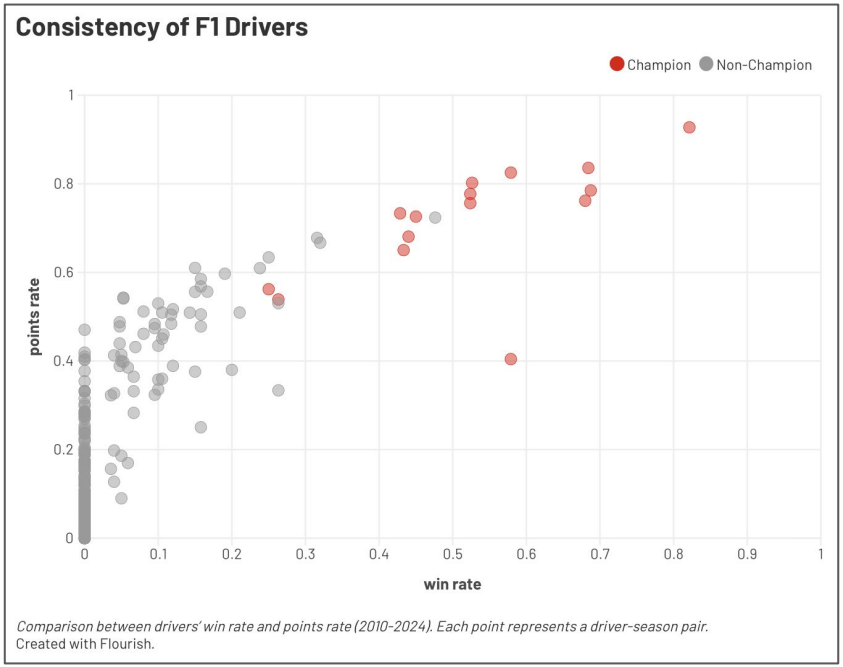


Consistency Wins Championships

SQ2: Do F1 world champions win more races?

This chart compares drivers' win rate[1] and points rate[2], with world champions highlighted in red and non-champions shown in grey. Races before 2010 are excluded, as the scoring system changed significantly that year. Both metrics are normalized to allow comparison across seasons with slightly different scoring systems.

World champions tend to appear in the right area of the chart, indicating a high number of victories. Non-champions are mostly concentrated in the left area, reflecting lower win rates. However, the presence of both champions and non-champions in the central area shows that winning the championship does not necessarily require extreme dominance: some champions secured titles through consistency rather than frequent victories, while some non-champions achieved multiple wins but were narrowly beaten over the course of the season.



Comparison between drivers' win rate and points rate (2010-2024). Each point represents a driver-season pair.
Created with Flourish.

[1] wins divided by total races

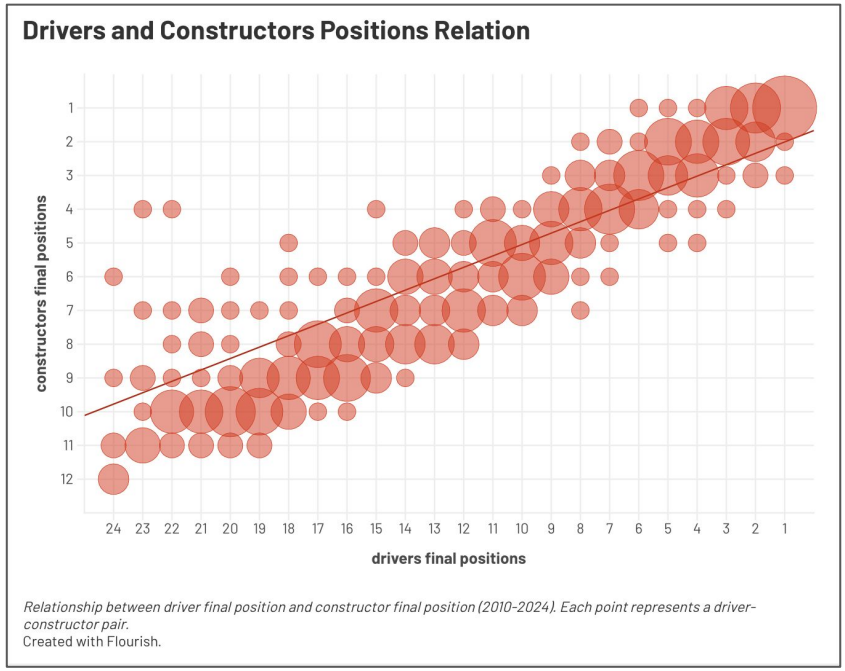
[2] points achieved by the drivers divided by the maximum points achievable in a season

A Winning Car Is Not Enough

SQ3: Does driving for the best team guarantee a world championship?

This chart compares a driver's final championship position with their constructor's final position. Each bubble represents a driver-constructor combination, with larger bubbles indicating combinations that occurred multiple times across different seasons. The analysis includes only seasons from 2010 onward, as fewer drivers and constructors competed in earlier periods.

The strong diagonal pattern shows a clear relationship between team performance and driver results: drivers in top-performing teams generally finish higher in the championship. However, the chart also reveals that driving for the best team does not automatically guarantee a world title: there are several cases where drivers from leading teams finish lower.



Relationship between driver final position and constructor final position (2010-2024). Each point represents a driver-constructor pair.
Created with Flourish.

Overall, the visualization suggests that a competitive car is a crucial advantage, but individual performance remains a decisive factor in winning the championship.

World Champions Are Built on Multiple Strengths

RQ: What makes an F1 driver a World Champion?

Becoming a Formula 1 World Champion is the result of a **combination of factors**, rather than a single dominant variable.

- **Elite performance:** Only a handful of drivers make it to the highest level and tend to dominate for years, highlighting the extraordinary skill and dedication needed to excel.
- **Career longevity:** Constant participation over many seasons provides a different perspective on success, showing that remaining competitive for a long time can turn a driver into a legend.
- **Geographical advantage:** Most champions come from Europe, particularly Great Britain, highlighting the importance of a strong racing culture and technical heart in shaping top drivers.
- **Consistency on track:** While winning races is important, consistent point scoring across a season is often just as crucial for securing a championship.
- **Team performance:** Driving for a top team provides a significant advantage, but a competitive car alone does not guarantee a title.

So, becoming a Formula 1 World Champion requires **not just talent**, but a combination of skill, perseverance, culture, reliability, and collaboration.

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